UC Irvine School of Education

Lesson Plan: Describing Polygons

Part 1: Classroom Information			
Grade: 3	Group Size: 32	Content Area: Math	Lesson Length: 60 min
A. Student Context (TPEs 1, 2, 4)			
	Needs	Strengths	Accommodations
Students with Special Needs (IEP and/or 504)	[removed for privacy]		
Students with Specific Language Needs (Multilingual Learners)			
Students with Other Learning Needs			
B. Building on Students' Assets (TPE 1)			
This lesson builds on students' prior knowledge of shape attributes to help them make sense of the characteristics of a polygon. Previous lessons focus on what closed/open shapes and plane shapes are. Students must rely on their funds of knowledge to effectively communicate with their peers and think about where they have seen polygons in their own lives.			

Part 2: Planning for the Lesson	
A. Standards (TPE 3)	
Key Content Standard	Math.3.G.A.1 **Understand that shapes in different categories (e.g., rhombuses, rectangles, and others) may share attributes (e.g., having four sides), and that the shared attributes can define a larger category (e.g., quadrilaterals).** Recognize rhombuses, rectangles, and squares as examples of quadrilaterals, and draw examples of quadrilaterals that do not belong to any of these subcategories Building towards: Math.3.MD.D.8 Solve real-world and mathematical problems involving perimeters of **polygons**, including finding the perimeter given the side lengths, finding an unknown side length, and exhibiting rectangles with the same perimeter and different areas or with the same area and different perimeters
B. Learning Goal/Objective (TPE 3)	
Students will compare and contrast attributes of shapes in different categories, identifying the required characteristics of polygons and other categories of shapes and determine appropriate labels for different shapes based on shared attributes.	

C. **Incorporating Academic Language** (TPE 7)	
Language Supports	
Function	Compare and contrast attributes Identify required characteristics Determine appropriate labels
Vocabulary/Symbols	Polygon Closed, open, plane/2-dimensional, line segment, curved, corner, cross Triangle, quadrilateral, pentagon, hexagon, octagon, decagon
Active Listening	Think-pair-share, open-ended questions
Grammatical Structures	Sentence frame: I know this is a polygon because it is ___, ___, ___, and ___.
Written, Visual, and Verbal Communication	Anchor chart
Language Objective	
Model: Students will [function] [academic content] by [language supports].	Students will compare and contrast categories of shapes using mathematical vocabulary Students will identify polygon attributes using mathematical vocabulary (plane, closed, line segments) Students will categorize polygons using mathematical vocabulary (triangle, quadrilateral, pentagon, hexagon, octagon, decagon)
Math Language Routines for Math Lessons (TPE 1, 3, 4, 7) Highlight these language supports in the body of your lesson in yellow.	
Mathematical language in this lesson is supported through discourse structures and a co-constructed anchor chart that students can refer to. MLR 2: Collect and Display I will use an anchor chart to display student ideas about polygons based on student discussion and share-outs MLR 8: Discussion Supports Student ideas from discussion/share-outs will be revoiced, and students will be asked probing/pressing questions to get them to think more deeply and elaborate	
Language Supports (TPE 1, 3, 4, 7) Highlight these language supports in the body of your lesson in yellow.	
Sentence frames, visual supports for vocabulary, modeling	

D. **Assessments** (TPE 5)	
Assessment Strategies	Evidence of Student Learning
Student discourse (think-pair-share, turn-and-talk)	Notice and wonder e.g.: “I think A B and C go together because they are made of straight lines” Polygon vs not a polygon e.g.: “The shapes on the polygon side are all made of straight lines” Students are conversing with their peers and show body language that indicates that they are actively listening to each other
Questioning	“One shared attribute is…” “A ______ has ______ sides and ______ angles.”
Exit Ticket	Students will draw a closed, 2D/plane shape with 6 sides made up of straight line segments that do not cross and meet at corners. Students will identify their shape as a hexagon and explain the requirements that their shape meets to be classified as a polygon. e.g. “I know this is a polygon because it is closed, 2-dimensional, made of straight lines, and the sides meet at corners.”

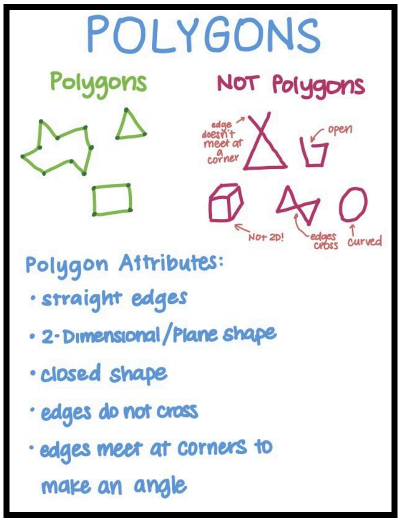
Part 3: Instructional Sequence - Engaging Students in the Learning Process	
Language support and/or discourse structures Checking for Understanding strategies - how and when are these happening Supports for students with specific learning needs	
Warmup/ Opener (5 mins)	Establish partners and partner discussion norms → have students move to the carpet. Make sure students are sitting where they will be able to see the board. Review: What is an attribute? → turn and talk Which 3 go together - review what students should be doing during this exercise - silent think time → share with a partner [monitor] → class discussion [write student ideas on board] possible answers: - ABC: straight lines - ABD: 2-dimensional - ACD: lines do not cross - BCD: shape is closed Review vocabulary that surfaces during discussion by asking students to clarify meaning (e.g. “What do you mean by 2-dimensional?”) Choral reading of learning goal: I can identify the required attributes of a polygon.

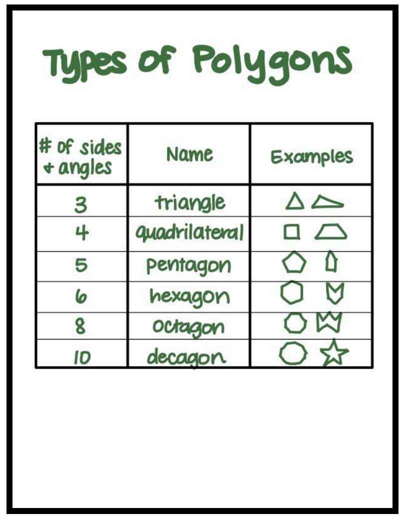
Body (40 mins)	Polygons:
	
	Open-ended question - What is a polygon? - have students make predictions
	- think-pair-share. Revoice student ideas.
	
	Display examples and non-examples of polygons
	- "The shapes on this side [point to the left side of screen] are polygons. The shapes on this side [point to right side of screen] are not. I want you to think to yourself about what attributes the polygons have that the non-examples do not have. When you have thought of an attribute, give me a silent thumbs up on your chest."
	    - Once the majority of students have thumbs up, tell students they have 1 minute to discuss their attribute with their partner [listen to student conversations]
	    - call on students to share based on conversations. Display student ideas on anchor chart
	
	Categories of polygons:
	
	Display slide with Shape Attributes anchor chart → display slide with zoom in on Number of Sides
	- think-pair-share: "Discuss with your partner: What do these words mean? How do you know when to use these words?" [monitor]
	- final understanding should be that the names are determined by the number of sides/angles in the shape
	- add these to anchor chart as examples
	Display slide with quadrilateral from last exit ticket
	- "Is this a quadrilateral? How do you know?"
	- Volunteer share. Point out sides and angles on display.
	Display slide with hexagons
	- "Are these hexagons? How do you know?"
	- think-pair-share. Point out sides and angles on display.
	Display slide with octagons
	- "These polygons are all octagons. Can everyone say 'octagon'? [choral repeat] What do you think is a shared attribute of octagons?"
	- think-pair-share. Point out sides and angles on display
	- add to anchor chart
	Display slide with decagons
	- "These polygons are all decagons. Can everyone say 'decagon'? [choral repeat] What do you think is a shared attribute of decagons?"
	- think-pair-share. Point out sides and angles on display
	- add to anchor chart
	
	Partner activity instructions:
	- have students read out instructions on slide
	- clarify each step after student reads
	- Model process
	- Give students quarter sheet of paper and have them go back to their desks to do activity
	- give students 8 minutes to complete activity (check in and adjust time accordingly). Monitor student description of attributes

<table>
<tr><td></td><td>Class discussion: What attribute do you think helped you the most in matching your drawings?</td></tr>
<tr><td>Closure (15 mins)</td><td>Choral read learning goal

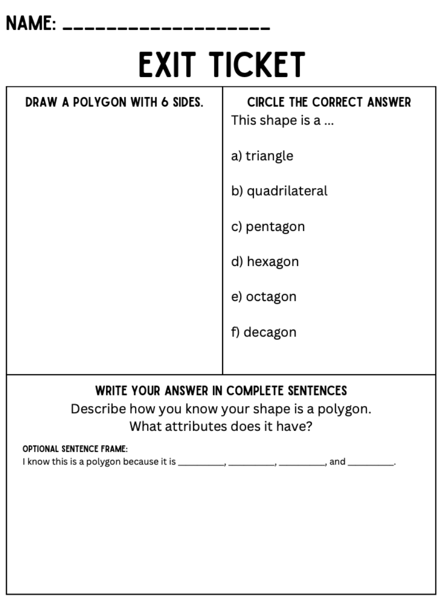
What are the required attributes of polygons?
What attributes do all pentagons share? Octagons?

Exit ticket: show students exit ticket and read out instructions. Have students pass out</td></tr>
<tr><td>Resources/ Materials</td><td><u>Teacher Materials</u>: Chart paper, chart markers, chart drafts, <u>Slides</u>





<u>Student Materials</u>: Blank quarter sheet of paper, pencils, exit Tickets


NAME: ___________________

EXIT TICKET

DRAW A POLYGON WITH 6 SIDES.

CIRCLE THE CORRECT ANSWER
This shape is a ...
a) triangle
b) quadrilateral
c) pentagon
d) hexagon
e) octagon
f) decagon

WRITE YOUR ANSWER IN COMPLETE SENTENCES
Describe how you know your shape is a polygon.
What attributes does it have?

OPTIONAL SENTENCE FRAME:
I know this is a polygon because it is ________, ________, ________, and ________.
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